Project Topic

Your Name

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1 Introduction

The introduction will broadly introduce your topic, motivate it, and describe what will be done. Traffic management remains a crucial issue in large cities, especially rapidly developing cities without extensive urban planning. Poor traffic management results in congestion, unnecessary emissions, and accidents that impact the safety of the city's residents. In order to manage the flow of traffic within a city, many cities turn to data-driven approaches that inform urban planning.

"Smart cities" refer to cities that utilize data and computational techniques to improve the functioning of a city. For example, a smart city may use location services on cell phones to determine the best times and places to collect waste within a city. Intelligent traffic systems, a crucial component of smart cities, aggregate and analyze data to manage traffic within a city. Intelligent traffic systems may make use of traffic data, such as data gathered from Uber or from sensors placed around the city, to model traffic in real time. These traffic models can then be used to create software that responds to current traffic flows and decreases congestion, such as software like Google Maps that re-routes traffic to less congested areas. Urban planners, public sector transportation employees, politicians, and everyday people rely on accurate traffic models to make informed decisions about new transit routes or traffic policy.

Traffic accident analysis is an important subsection of traffic management.

The majority of traffic accident prediction studies focus on determining merely whether or not an accident will happen, but leave out additional analysis crucial to urban planners or policymakers that help these people to make informed traffic planning decisions. Traffic accident models which solely predict accidents do not explain how or what factors led into the algorithm deciding the result, which is both uninformative and seen as unreliable by policymakers who don't understand how the result came to be. Thus, many traffic studies rely on explanatory models to analyze the underlying factors that contribute to an accident or the severity of the accident.

This study aims to provide better insight into traffic accidents in small American towns, including factors that can influence the likelihood and severity of a traffic accident. Using XGBoost analysis, this study will analyze a dataset of 10,000 traffic accidents in Wooster, Ohio over the past five years in order to model the likelihood of accidents and determine which factors most contributed to causing a traffic accident. Severity of the traffic accident will additionally be considered.

many smart cities utilize road sensors to detect car movements at specific times and locations and use this data to inform the creation of optimal transit routes within the city.

2 Background

The field has historically relied on statistical regression analysis in order to make decisions regarding traffic safety, such as ARIMA or linear regression techniques. Recent approaches using machine learning and neural networks produce models of traffic flows, jams, and accidents have shown a marked increase in accurately modeling traffic. Signal traffic control is another field of study in which machine learning analysis is used to decrease congestion in urban settings.

Many studies have observed the superior performance of decision tree networks such as random forest and XGBoost over neural networks such as LSTM or CNNs. Urbanized cities in North America have more data available to train machine learning models or neural networks, such as Uber data, sensors, cameras, and satellite data. Developing cities face greater challenges in acquiring the data required to model traffic flows and often require collaboration with local governments in order to place sensors or otherwise acquire traffic data that can be used for analysis.

3 Related Works

4 Methodology

Dijkstra's algorithm was chosen for single-source shortest path computation on graphs with non-negative edge weights, following the implementation presented in CLRS [**?**].

5 Results

An exemplary figure is shown in Figure 1. Any time a figure is used, it must have a capation and be directly referred to and discussed in the text.

6 Conclusion

References

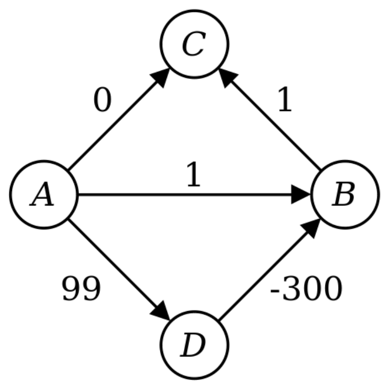

Figure 1: Dijkstra's algorithm does not work on graphs with negative edge weights, as evidenced here.